ICML
International Conference On Machine Learning

Project Page

Evaluating Object-Centric Models beyond Object Discovery

Krishnakant Singh, Simone Schaub-Meyer, Stefan Roth

visual inference

TECHNISCHE
UNIVERSITÄT
DARMSTADT

VLM-based evaluation protocol

VLM-based evaluation: An ideal eval. protocol

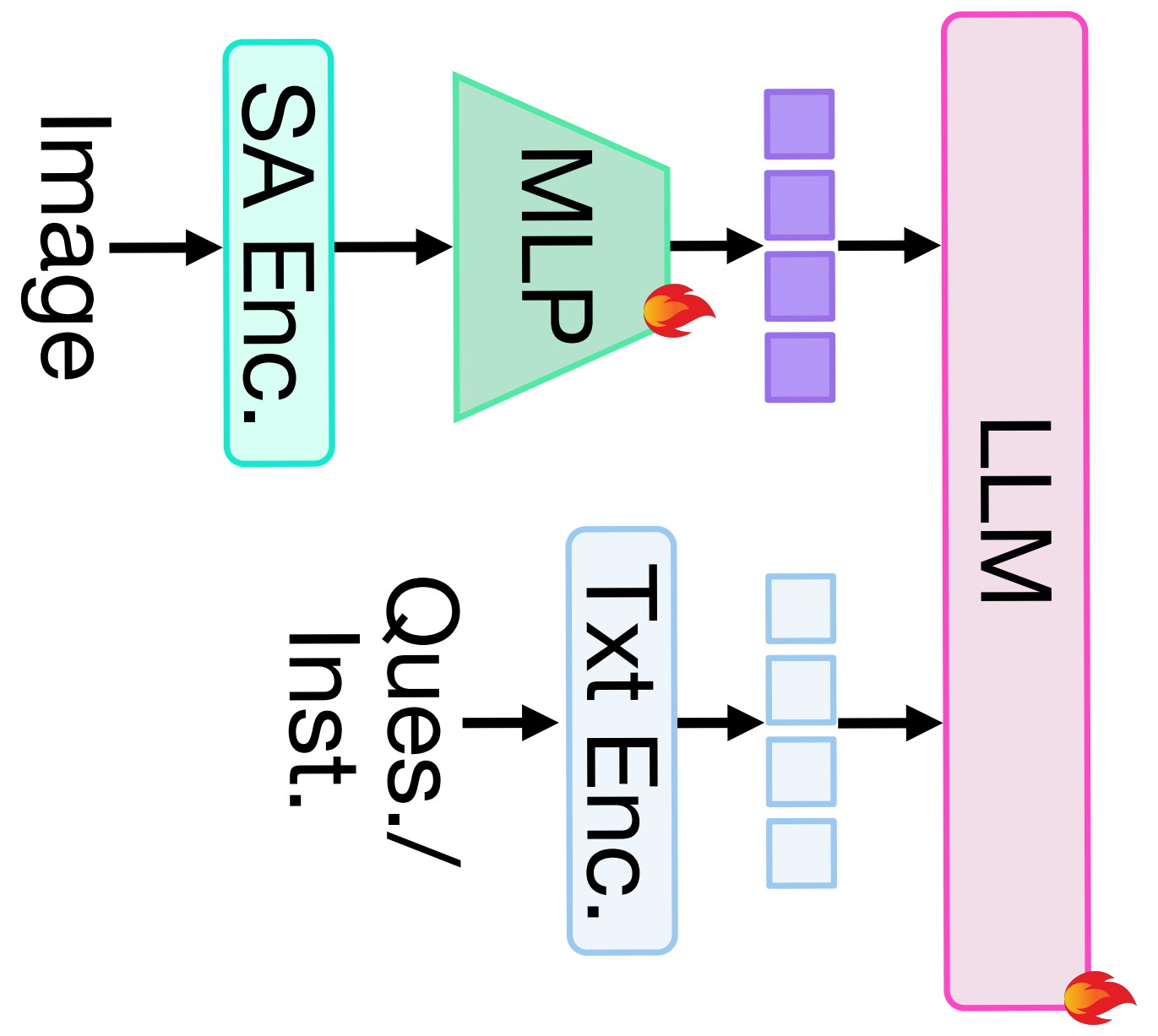
- Near-zero amortized cost
- Evaluate diverse properties

Out-of-distribution robustness

How many people are there in the image if 2 more people are added?

Counterfactual reasoning

LLaVA[1] inspired evaluation framework



- Alignment stage trains only the MLP.
- Instruction-tuning stage trains the LLM and MLP together.
- VLM-based evaluation allows for benchmarking of diverse properties.

Visual question answering accuracy

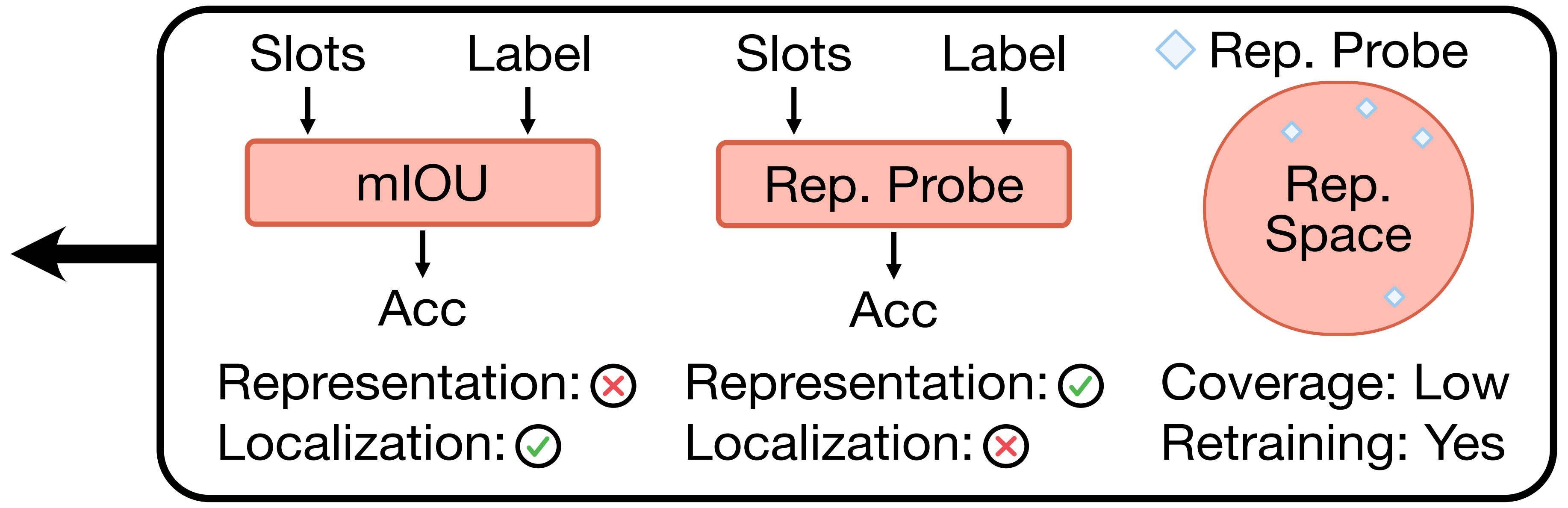
Model	GQA	POPE	VQAv2	CVQA	OODCV	SugarC.
DINOv2[2]	57.77	82.01	71.15	36.96	58.00	82.05
DINOSAURv2[3]	53.23	81.76	63.84	34.52	53.90	75.98
SPOT[4]	51.06	79.74	60.94	36.35	53.34	71.65
StableLSD[5]	51.45	81.51	62.06	38.26	52.89	72.92

Takeaway

VLM-based evaluation enables benchmarking diverse properties of object-centric latents without per-benchmark training.

Rethinking OCL evaluations

Issues with "what" and "where" probing



Representation: ⊗
Localization: ✓

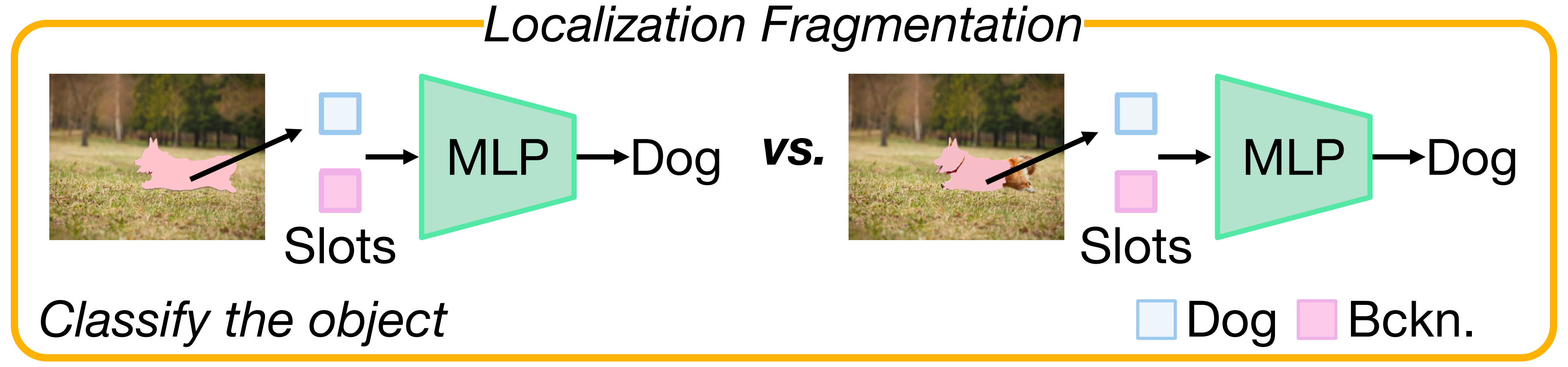
Representation: ✓
Localization: ⊗

Coverage: Low
Retraining: Yes

- Object discovery metrics such as mIoU only measure localization.
- Representation probing of OCL methods requires re-training for every property and cannot evaluate diverse properties.

Issues with disjoint evaluation

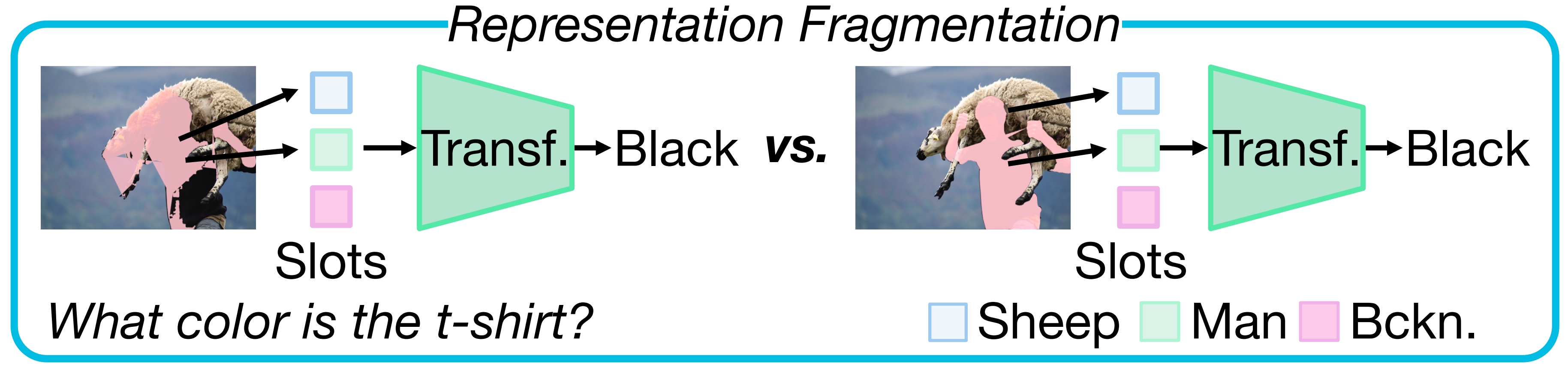
Localization Fragmentation



Classify the object

☐ Dog ☐ Bckn.

Representation Fragmentation



What color is the t-shirt?

☐ Sheep ☐ Man ☐ Bckn.

AwGA metric

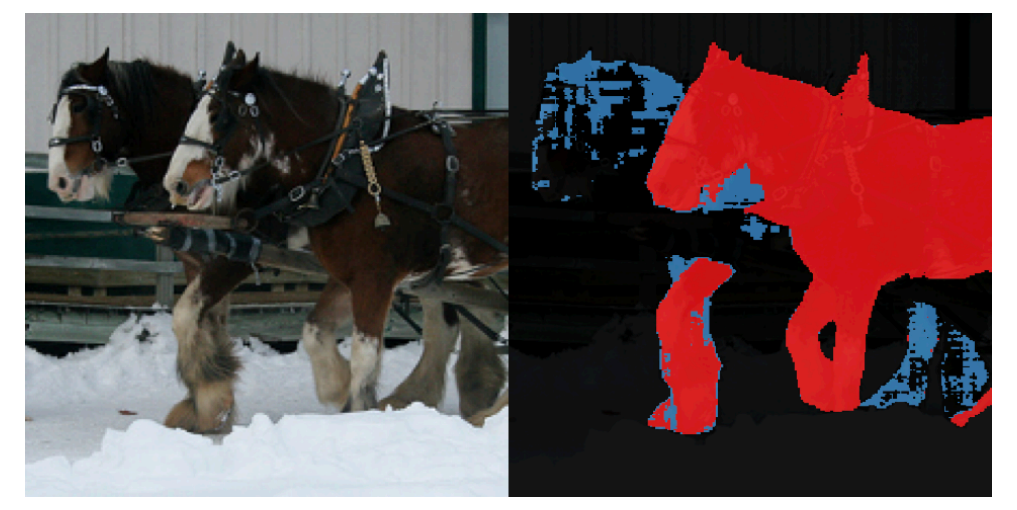
- Disjoint evaluation does not penalize models for fragmentation issues.

This work was funded by the ERC under EU Horizon 2020 Grant No. 866008, supported in part by the DFG under EXC-3057/1 "Reasonable Artificial Intelligence" Project No. 533677015, and SSM was funded by DFG Project No. 529680848.

A joint evaluation metric

Enhanced GQA dataset: A grounding dataset

What color is the uniform? A: Red



What kind of animal is pulling the cart? A: Horse

Do you see an umbrella above her? A: Yes

Attribution-aware grounded accuracy (AwGA)

$$\mathrm{AwGA} = \frac{1}{N}\sum_{i=1}^{N} \mathbb{1}(\hat{y} = y)\ \mathrm{mIoU}(\mathrm{TopK}(\mathcal{A}_{\mathrm{pred}}), \mathcal{G}_{\mathrm{GT}})$$

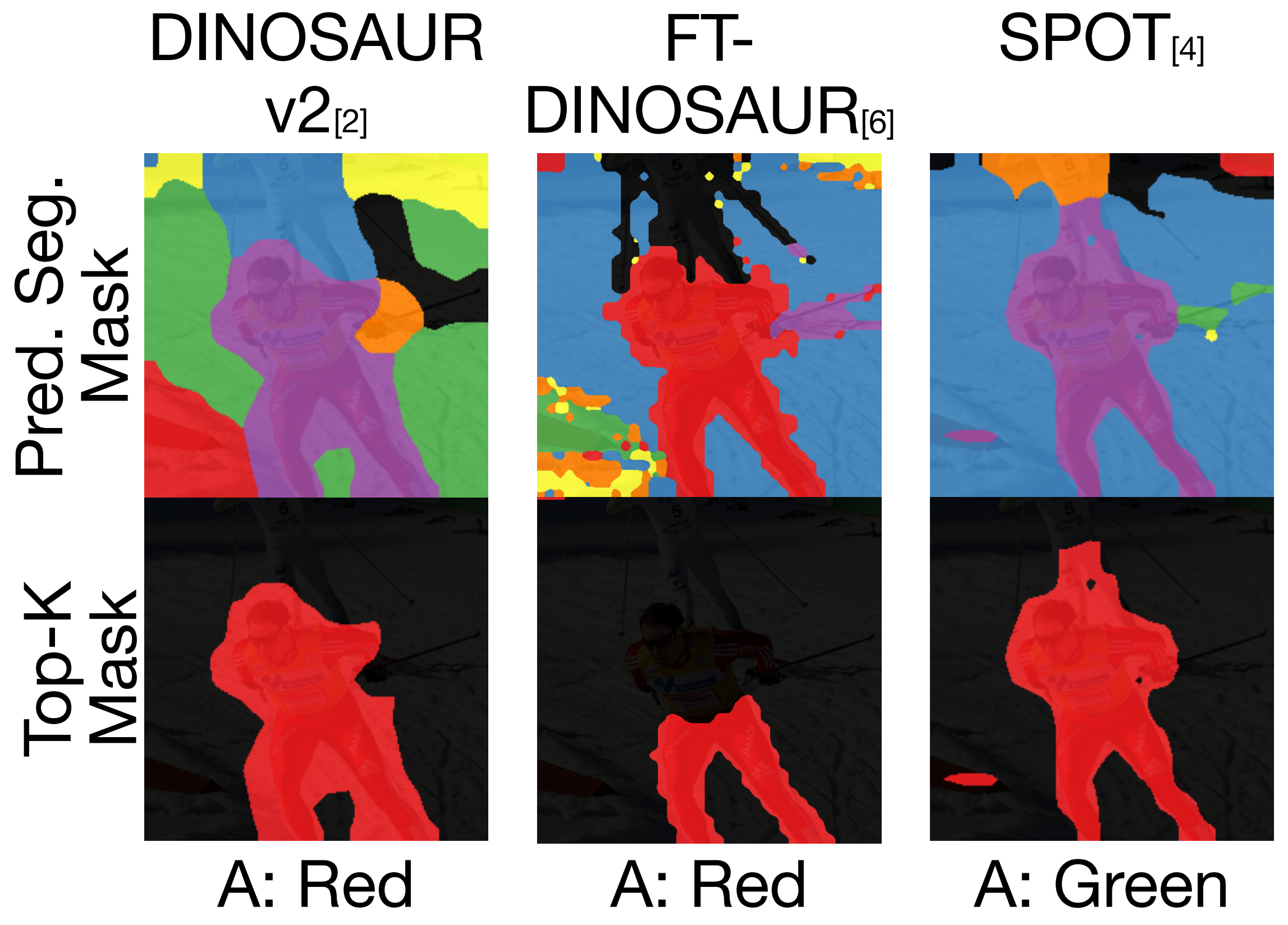


Image
G. Mask

What color is the uniform? Ans: Red

Model	mIOU	Acc.	G-Acc.	AwGA
DINOSAURv2[2]	47.99	66.27	32.40	11.92
FT-DINOSAUR[6]	59.09	61.05	33.94	13.25
SPOT[4]	53.76	64.08	38.45	12.81

Takeaway

AwGA provides a single metric that jointly captures how well object-centric latents localize objects and how useful they are for downstream tasks.

---

[1] Liu *et al.*, Visual Instruction Tuning, NeurIPS, 2023.
[2] Oquab *et al.*, DINOv2: Learning Robust Visual Features without Supervision, TMLR, 2024.
[3] Seitzer *et al.*, Bridging the Gap to Real-World Object-Centric Learning, ICLR, 2023.
[4] Kakogeorgiou *et al.*, SPOT: Self-Training with Patch-Order Permutation for OCL with Autoregressive Transf., CVPR, 2024.
[5] Didolkar *et al.*, FT-DINOSAUR: Zero-Shot Object-Centric Representation Learning, ICLR, 2025.
[6] Jiang *et al.*, LSD: Object-Centric Slot Diffusion, NeurIPS, 2023.